Chapter 7

Series

7.1 Series and their partial sums

In this chapter we study **series**: infinite sums of the form

$$a_0 + a_1 + a_2 + \cdots + a_n + \cdots = ?$$

Making sense of such infinite sums – deciding when they have a value, and what that value is – is one of the fundamental problems of analysis. Series will recur later in these notes, for instance when we connect them to improper integrals (Chapter 13).

Series and their partial sums

Given a sequence $\{a_n\}_{n\in\mathbb{N}}$ of real numbers, we form the **partial sums**

$$s_n = a_0 + a_1 + a_2 + \cdots + a_n = \sum_{k=0}^{n} a_k.$$

The **series** (or infinite sum) associated to $\{a_n\}_{n\in\mathbb{N}}$ is defined as the limit of the partial sums:

$$\sum_{k=0}^{\infty} a_k = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \sum_{k=0}^{n} a_k.$$

If the limit exists and is finite, we say the series *converges*; if the limit is infinite, the series *diverges*; if the limit does not exist, the series is *indeterminate*.

7.2 Geometric series

One of the most important examples is the geometric series.

Geometric series

For $r \in \mathbb{R}$, the geometric series is

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \cdots .$$

Its partial sums are given by the finite geometric sum formula (Chapter 0):

$$s_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r} \quad \text{for } r \neq 1.$$

Taking the limit as $n \to \infty$, we obtain:

$$\sum_{n=0}^{\infty} r^n = \begin{cases} \frac{1}{1-r} & \text{if } |r| < 1 \quad \text{(the series converges)}, \\ +\infty & \text{if } r \geq 1 \quad \text{(the series diverges)}, \\ \text{does not exist} & \text{if } r \leq -1 \quad \text{(the series is indeterminate)}. \end{cases}$$

(For $r = 1$ the formula for s_n does not apply, but directly $s_n = n + 1 \to +\infty$. For $r = -1$ the partial sums alternate: $s_n = 1, 0, 1, 0, \ldots$)

Zeno's paradoxes

Achilles, the fastest runner of Greek legend, races a tortoise that starts 100 *meters ahead. By the time Achilles reaches the tortoise's starting point, it has crawled a little further; by the time he covers that gap, it has moved on again; and so on forever. There are infinitely many catch-up stages, so it seems Achilles can never overtake the tortoise. A related paradox, the* dichotomy*, says that to reach a wall you must first cover half the distance, then half of what remains, and so on, so you never arrive. The geometric series resolves the mathematical puzzle: infinitely many stages can add up to a finite total. The wall is reached after* $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$*, the whole distance. If Achilles runs ten times faster than the tortoise, the catch-up distances are* $100 + 10 + 1 + \frac{1}{10} + \cdots = \frac{100}{1 - \frac{1}{10}} = 111.\overline{1}$ *meters, and after that Achilles is in front.*

These puzzles go back to Zeno of Elea (c. 490–425 BC), from a Greek city in southern Italy. He was a pupil of the philosopher Parmenides, who taught that motion is an illusion, and devised the paradoxes to defend him. His book is lost; we know the paradoxes mainly through Aristotle.

7.3 The Cauchy Condensation Test and p-series

p-Series

For $p > 0$, a **p-series** is the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. In the case $p = 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is called the **harmonic series**.

To determine exactly when a p-series converges, we introduce a new comparison tool that is especially well-suited to decreasing sequences.

Theorem 7.1 (Cauchy Condensation Test)**:** Let $\{a_n\}_{n\in\mathbb{N}_+}$ be non-negative and decreasing (i.e. $a_1 \geq a_2 \geq \cdots \geq 0$). Then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \qquad \Leftrightarrow \qquad \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Proof. Since $a_n \geq 0$, both partial sums

$$s_n = \sum_{k=1}^{n} a_k \qquad \text{and} \qquad t_m = \sum_{k=0}^{m} 2^k a_{2^k}$$

are monotone increasing (in n and m respectively), so by the Monotone Convergence Theorem (Theorem 5.1), $\sum a_n$ converges if and only if $\{s_n\}$ is bounded above, and $\sum 2^k a_{2^k}$ converges if and only if $\{t_m\}$ is bounded above. We show that $\{s_n\}$ is bounded if and only if $\{t_m\}$ is bounded.

(t_m bounded $\Rightarrow$ s_n bounded). Fix n and choose m such that $n < 2^{m+1}$. Grouping the terms of s_n into blocks $a_{2^k}, a_{2^k+1}, \ldots, a_{2^{k+1}-1}$ (each of length 2^k), and using that $\{a_n\}$ is decreasing (so every term of the kth block is $\leq a_{2^k}$), we get

$$s_n \leq a_1 + \sum_{k=1}^{m} \left(a_{2^k} + \cdots + a_{2^{k+1}-1}\right) \leq a_1 + \sum_{k=1}^{m} 2^k a_{2^k} = t_m.$$

So if $\{t_m\}$ is bounded above, so is $\{s_n\}$.

(s_n bounded $\Rightarrow$ t_m bounded). Similarly, grouping the terms of s_{2^m} into blocks $a_{2^{k-1}+1}, \ldots, a_{2^k}$ (each of length 2^{k-1}), and using that every term of this block is $\geq a_{2^k}$, we get

$$s_{2^m} \geq a_1 + a_2 + \sum_{k=2}^{m} 2^{k-1} a_{2^k} = a_1 + \frac{1}{2}\sum_{k=1}^{m} 2^k a_{2^k} = a_1 + \frac{1}{2}\left(t_m - a_1\right) = \frac{1}{2}\left(a_1 + t_m\right),$$

so that $t_m \leq 2s_{2^m} - a_1 \leq 2s_{2^m}$. Since $\{s_n\}$ is increasing, $s_{2^m} \leq \sup_n s_n$, so if $\{s_n\}$ is bounded above, so is $\{t_m\}$.

Combining both directions proves the theorem. □

Corollary 7.2 (Convergence of p-series)**:** For $p > 0$, the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $0 < p \leq 1$. In particular (the case $p = 1$), the **harmonic series** $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Proof. The sequence $a_n = \frac{1}{n^p}$ is non-negative and decreasing, so the Cauchy Condensation Test (Theorem 7.1) applies:

$$2^k a_{2^k} = \frac{2^k}{2^{kp}} = \left(2^{1-p}\right)^k,$$

so $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is a geometric series with ratio $r = 2^{1-p} > 0$. By the Geometric Series remark above, this converges if and only if $r < 1$, i.e. $2^{1-p} < 1$, i.e. $p > 1$. By the Cauchy Condensation Test, the original series $\sum \frac{1}{n^p}$ converges if and only if $p > 1$.

For $p = 1$ we get $a_n = \frac{1}{n}$, the harmonic series, which by the above diverges. □

7.4 Necessary condition

Necessary condition for convergence

If $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Proof. Let $s_n = \sum_{k=0}^{n} a_k$ and $s = \lim_{n\to\infty} s_n$. Then

$$a_n = s_n - s_{n-1} \to s - s = 0 \quad \text{as } n \to \infty.$$

□

Remark: The converse is *false*! For example, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (Corollary 7.2) even though $\frac{1}{n} \to 0$.

On the other hand, the necessary condition settles the p-series with $p \leq 0$: then $\frac{1}{n^p} = n^{-p} \geq 1$ does not tend to 0, so $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges. Together with Corollary 7.2: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

7.5 Convergence tests for series with non-negative terms

For series with $a_n \geq 0$ for all n, the sequence of partial sums $\{s_n\}$ is non-decreasing, so it either converges to a finite limit or diverges to $+\infty$.

Comparison Test

Let $\{a_n\}$ and $\{b_n\}$ be sequences with $0 \leq a_n \leq b_n$ for all n. Then:

- If $\sum_{n=0}^{\infty} b_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges.
- If $\sum_{n=0}^{\infty} a_n$ diverges, then $\sum_{n=0}^{\infty} b_n$ diverges.

Proof. Let $s_n = \sum_{k=0}^{n} a_k$ and $t_n = \sum_{k=0}^{n} b_k$. Both sequences are increasing (the terms are non-negative), and $s_n \leq t_n$ for all n. If $\sum_{n=0}^{\infty} b_n$ converges, then $\{t_n\}$ is bounded above (by its limit), hence so is $\{s_n\}$, and $\{s_n\}$ converges by the Monotone Convergence Theorem (Theorem 5.1). The second claim follows from the first: if $\sum_{n=0}^{\infty} b_n$ converged, then so would $\sum_{n=0}^{\infty} a_n$. □

Example 7.1: The p-series with $p = 2$, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges because $\frac{1}{n^2} \leq \frac{2}{n(n+1)} = 2\left(\frac{1}{n} - \frac{1}{n+1}\right)$ and

$$\sum_{n=1}^{N} 2\left(\frac{1}{n} - \frac{1}{n+1}\right) = 2\left(1 - \frac{1}{N+1}\right) \to 2.$$

In practice it is often easier to compare the *behavior* of a_n and b_n for large n than to find an explicit inequality between them:

Asymptotic Comparison Test

Let $\{a_n\}$ and $\{b_n\}$ be sequences with $a_n > 0$ and $b_n > 0$ for all n, and suppose that

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L \qquad \text{with } 0 < L < +\infty.$$

Then $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ either both converge or both diverge.

Proof. Let $s_n = \sum_{k=0}^{n} a_k$ and $t_n = \sum_{k=0}^{n} b_k$. Both are increasing, so each of them either converges or diverges to $+\infty$, according to whether it is bounded or not. Since $\frac{a_n}{b_n} \to L > 0$, there exists $N \in \mathbb{N}$ such that $\frac{L}{2} \le \frac{a_n}{b_n} \le 2L$, that is

$$\frac{L}{2}\, b_n \le a_n \le 2L\, b_n, \qquad \text{for all } n > N.$$

Summing the right inequality from $N+1$ to n gives $s_n - s_N \le 2L(t_n - t_N) \le 2Lt_n$, so $s_n \le s_N + 2Lt_n$ for all $n > N$. Similarly, the left inequality gives $t_n \le t_N + \frac{2}{L} s_n$. Hence $\{s_n\}$ is bounded if and only if $\{t_n\}$ is bounded, which proves the claim. □

Example 7.2: Consider $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$. For large n the terms behave like $\frac{n}{n^3} = \frac{1}{n^2}$, so we take $b_n = \frac{1}{n^2}$:

$$\frac{a_n}{b_n} = \frac{n^2(n+1)}{n^3+2} = \frac{n^3+n^2}{n^3+2} \to 1.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (Corollary 7.2), so does $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$.

Example 7.3: Consider $\sum_{n=1}^{\infty} \sin\frac{1}{n}$. Its terms are positive, since $0 < \frac{1}{n} \le 1 < \pi$. Take $b_n = \frac{1}{n}$. Since $\frac{1}{n} \to 0$, $\frac{1}{n} \neq 0$ and $\lim_{x\to 0} \frac{\sin x}{x} = 1$, the Substitution Theorem for sequences (Theorem 6.15, item 7(b)) gives

$$\frac{a_n}{b_n} = \frac{\sin\frac{1}{n}}{\frac{1}{n}} \to 1.$$

Since the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, so does $\sum_{n=1}^{\infty} \sin\frac{1}{n}$.

Ratio Test (d'Alembert's test)

Let $\sum_{n=0}^{\infty} a_n$ be a series with $a_n > 0$ for all n, and let

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}.$$

Then:

- If $L < 1$, the series converges.
- If $L > 1$, the series diverges.
- If $L = 1$, the test is inconclusive.

Proof. Let $s_n = \sum_{k=0}^{n} a_k$. Suppose $L < 1$ and fix q with $L < q < 1$. Since $\frac{a_{n+1}}{a_n} \to L < q$, there exists $N \in \mathbb{N}$ such that $a_{n+1} \leq q\, a_n$ for all $n \geq N$, hence $a_n \leq a_N q^{n-N}$ for all $n \geq N$. Therefore, for $n > N$,

$$s_n \leq s_N + a_N \sum_{k=1}^{n-N} q^k \leq s_N + \frac{a_N}{1-q},$$

so the increasing sequence $\{s_n\}$ is bounded above, and the series converges. If $L > 1$, the Ratio Test for sequences (Chapter 6) gives $a_n \to +\infty$; in particular $a_n \not\to 0$, so by the necessary condition the series does not converge, that is, it diverges. Finally, $\sum_{n=1}^{\infty} \frac{1}{n}$ (divergent) and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ (convergent) both have $L = 1$, so no conclusion is possible in that case. □

Example 7.4: For $\sum_{n=1}^{\infty} \frac{n}{2^n}$, we have

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)/2^{n+1}}{n/2^n} = \frac{n+1}{2n} \to \frac{1}{2} < 1,$$

so the series converges by the Ratio Test.

Root Test (Cauchy's test)

Let $\sum_{n=0}^{\infty} a_n$ be a series with $a_n \geq 0$ for all n, and let

$$L = \lim_{n\to\infty} \sqrt[n]{a_n}.$$

Then:

- If $L < 1$, the series converges.
- If $L > 1$, the series diverges.
- If $L = 1$, the test is inconclusive.

Proof. Let $s_n = \sum_{k=0}^{n} a_k$. Suppose $L < 1$ and fix q with $L < q < 1$. There exists $N \in \mathbb{N}$ such that $\sqrt[n]{a_n} \leq q$, that is $a_n \leq q^n$, for all $n > N$. Therefore, for $n > N$,

$$s_n \leq s_N + \sum_{k=N+1}^{n} q^k \leq s_N + \frac{1}{1-q},$$

so the increasing sequence $\{s_n\}$ is bounded above, and the series converges. If $L > 1$, then $\sqrt[n]{a_n} > 1$, that is $a_n > 1$, for all large n; in particular $a_n \not\to 0$, so by the necessary condition the series diverges. □

Example 7.5: For $\sum_{n=1}^{\infty} \left(\frac{n}{3n+1}\right)^n$, we have

$$\sqrt[n]{a_n} = \frac{n}{3n+1} \to \frac{1}{3} < 1,$$

so the series converges by the Root Test.

7.6 Alternating series

A series is called **alternating** if its terms alternate in sign.

Leibniz's Alternating Series Test

Let $\{a_n\}$ be a sequence such that:

1. $a_n \geq 0$ for all n,
2. $\{a_n\}$ is decreasing: $a_{n+1} \leq a_n$ for all n,
3. $\lim_{n\to\infty} a_n = 0$.

Then the alternating series $\sum_{n=0}^{\infty}(-1)^n a_n$ converges. Moreover, if $S = \sum_{n=0}^{\infty}(-1)^n a_n$ and $s_N = \sum_{n=0}^{N}(-1)^n a_n$, then the error satisfies:

$$|S - s_N| \leq a_{N+1}.$$

Proof. Let $s_N = \sum_{n=0}^{N}(-1)^n a_n$. Since $\{a_n\}$ is decreasing and non-negative,

$$\begin{aligned} s_{2m+2} &= s_{2m} - (a_{2m+1} - a_{2m+2}) \leq s_{2m}, \\ s_{2m+3} &= s_{2m+1} + (a_{2m+2} - a_{2m+3}) \geq s_{2m+1}, \\ s_{2m+1} &= s_{2m} - a_{2m+1} \leq s_{2m}. \end{aligned}$$

Hence

$$s_1 \leq s_3 \leq \cdots \leq s_{2m+1} \leq s_{2m} \leq \cdots \leq s_2 \leq s_0.$$

The even partial sums are decreasing and bounded below by s_1, and the odd ones are increasing and bounded above by s_0, so both converge by the Monotone Convergence Theorem (Theorem 5.1). Since $s_{2m} - s_{2m+1} = a_{2m+1} \to 0$, they have the same limit S. Therefore $s_N \to S$: given $\varepsilon > 0$, both $|s_{2m} - S|$ and $|s_{2m+1} - S|$ are $< \varepsilon$ for all large m.

Finally, S lies below every even partial sum and above every odd one, so it lies between any two consecutive partial sums s_N and s_{N+1}. Hence $|S - s_N| \leq |s_{N+1} - s_N| = a_{N+1}$. □

Example 7.6: The alternating harmonic series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n}$ converges by Leibniz's test, since $\frac{1}{n}$ is positive, decreasing, and tends to 0.

7.7 Absolute and conditional convergence

Absolute Convergence Test

If $\sum_{n=0}^{\infty}|a_n|$ converges, then $\sum_{n=0}^{\infty} a_n$ converges. In this case, we say the series **converges absolutely** and we have

$$\left|\sum_{n=0}^{\infty} a_n\right| \leq \sum_{n=0}^{\infty}|a_n|.$$

Proof. Let $s_n = \sum_{k=0}^{n} a_k$ and $t_n = \sum_{k=0}^{n} |a_k|$ be the partial sums of $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} |a_n|$, respectively. We split each term into its positive and negative parts:

$$a_n^+ = \max\{a_n, 0\}, \qquad a_n^- = \max\{-a_n, 0\},$$

so that

$$a_n = a_n^+ - a_n^- \qquad \text{and} \qquad 0 \le a_n^{\pm} \le |a_n|.$$

Since $\sum_{n=0}^{\infty} |a_n|$ converges, the Comparison Test shows that both $\sum_{n=0}^{\infty} a_n^+$ and $\sum_{n=0}^{\infty} a_n^-$ converge. Call their partial sums P_n and M_n, and their sums P and M. Then $s_n = P_n - M_n$, so by the algebra of limits for sequences (Theorem 6.15, item 6) $s_n \to P - M$, that is, $\sum_{n=0}^{\infty} a_n$ converges.

For the inequality: by the triangle inequality, $|s_n| \le t_n$ for all n. As $n \to \infty$ we have $|s_n| \to \left|\sum_{n=0}^{\infty} a_n\right|$ (the absolute value is continuous; Theorem 6.15, item 7(a)) and $t_n \to \sum_{n=0}^{\infty} |a_n|$, so the comparison of limits (Theorem 6.15, item 4) gives $\left|\sum_{n=0}^{\infty} a_n\right| \le \sum_{n=0}^{\infty} |a_n|$. □

Example 7.7: The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ converges absolutely because

$$\sum_{n=1}^{\infty} \left|\frac{(-1)^n}{n^2}\right| = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges (p-series with $p = 2 > 1$).

Remark: The Ratio and Root Tests also apply to series whose terms have any sign, through $\sum_{n=0}^{\infty} |a_n|$ (for the Ratio Test, assume $a_n \neq 0$ for all n). If $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} < 1$ or $\lim_{n\to\infty} \sqrt[n]{|a_n|} < 1$, then $\sum_{n=0}^{\infty} a_n$ converges absolutely, hence converges. If either limit is > 1, then $|a_n| \not\to 0$ (as in the proofs above), so $a_n \not\to 0$ and $\sum_{n=0}^{\infty} a_n$ does not converge.

Conditional Convergence

A series that converges but does not converge absolutely is said to **converge conditionally**. Absolute convergence is stronger than convergence (by the test above), and the difference matters when we change the order of the terms. A **rearrangement** of $\sum_{n=0}^{\infty} a_n$ is a series $\sum_{n=0}^{\infty} a_{\sigma(n)}$ with the same terms taken in a different order, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a bijection.

- *Absolutely convergent* series can be rearranged without changing the sum.
- *Conditionally convergent* series can be rearranged to converge to any real number or even diverge (Riemann rearrangement theorem).

Example 7.8: The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges conditionally but not absolutely, since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.